

Julien Boulanger

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I am currently a post-doc researcher at the *Center for mathematical modeling, University of Chile*, under the supervision of Rodolfo GUTIÉRREZ-ROMO. I am interested in geometric and dynamical properties of **translation surfaces**, **interval exchange transformations** but also other topics such as the study of **pseudo-Anosov diffeomorphisms** and **combinatorics on words**.

RESEARCH AND EDUCATION

Since September 2024 : Post-doctoral researcher, *Center for mathematical modeling, University of Chile*.

September 2021 to August 2024: PhD Thesis, *Institut Fourier*, Université Grenoble Alpes, supervised by Erwan LANNEAU and Daniel MASSART.

March to June 2024: Visiting researcher at the *Center for mathematical modeling, University of Chile*, as a PhD student.

2016-2021: Student at the *École normale supérieure de Paris*.

2020-2021: Preparation for the *Agrégation de mathématiques*. Université Grenoble Alpes.

2018-2019: Master 2 in pure mathematics. *Sorbonne université*, Paris. Master's thesis under the supervision of Erwan LANNEAU on Veech groups of translation surfaces.

February to June 2018: Internship at the University of Warwick under the supervision of Mark POLLICOTT on the ergodic theory of the horocycle flow on a hyperbolic surface.

2016-2018: Bachelor and Master 1 at the, *École normale supérieure*, Paris.

2014-2016: *Classe préparatoire aux grandes écoles*, MPSI, MP*. *Lycée du Parc*, Lyon.

2014: *Baccalauréat scientifique spécialité mathématiques*.

PUBLICATIONS

- JULIEN BOULANGER, *Connection points on regular polygons. To appear in the Journal of Modern Dynamics*, 2025.
- JULIEN BOULANGER, *Algebraic intersections, lengths, and Veech groups. To appear in the Annali della Scuola Normale Superiore di Pisa - Classe di Scienze*, 2025.
- JULIEN BOULANGER, ERWAN LANNEAU AND DANIEL MASSART, *Algebraic intersections in regular polygons. Annales Henri Lebesgue*, Volume 7 (2024), pp. 787-821
- JULIEN BOULANGER, *Lower bound for $KVol$ on the minimal stratum of translation surfaces. Geom. Dedicata* 218, 88 (2024).
- JULIEN BOULANGER AND SAM FREEDMAN, *There are no primitive Teichmüller curves in $Prym(2,2)$. Comptes Rendus Mathématiques*, Volume 362 (2024), pp. 167-170.
- JULIEN BOULANGER, *Central points of the double heptagon translation surface are not connexion points. Bulletin de la SMF*, Volume 2 (2022), pp. 459-472.

PREPRINTS

- JULIEN BOULANGER, RODOLFO GUTIÉRREZ-ROMO, ERWAN LANNEAU, *The Hurwitz problem for abelian differentials, Available on arXiv*, 2025
- JULIEN BOULANGER, *Algebraic interaction strength for translation surfaces with multiple singularities, Available on arXiv*, 2025
- JULIEN BOULANGER AND IRENE PASQUINELLI, *Algebraic intersections on Bouw-Möller surfaces, and more general convex polygons. Available on arXiv*, 2024.

TEACHING AND OUTREACH

2025: Instructor at the Departamento de Ingeniería Matemática, *Universidad de Chile*. Course on Ordinary Differential Equations, taught in spanish.

2021-2024: Instructor at the Département Licence sciences et technologies, *Université grenoble alpes*. Teaching service 64h/year. Several courses taught in french or in english.

2020-2021: Colles, MPSI, *Lycée Champollion*, Grenoble.

2019-2020: Volunteer with the association Mayaa Nepal, in Kathmandu ¹, 5 months during which I gave English and computer lessons to primary/middle school students.

2016-2017: Astrophysics teaching to high school students under the program *TalENS* of *pôle PESU* ².

ACADEMIC RESPONSIBILITIES

Since September 2025: Organizer of the bi-mensual Postdoc seminar of the *Center for Mathematical Modeling*.

2022-2024: Organizer of the weekly PhD seminar of the Institut Fourier. It includes the organisation of two two-days conferences (2022 and 2023) for the PhD students of the Institut Fourier.

TALKS

- Workshop of the ANR Adyct : "Aléatoire, Dynamique, spectre" in Peyresq, France, July 2025.
- Dynamics and Ergodic theory seminar, Paris Nord, June 2025.
- Geometry and groups seminar, Heidelberg, May 2025.
- Dynamical systems seminar, Montevideo, February 2025.
- Number theory seminar, Oregon State University, online. January 2025.
- Conference "Dyadisc 7: Brazilian, Chilean and French interplay for symbolic dynamics", Valparaíso, December 2024.
- Post-doc seminar of the center for mathematical modeling, Santiago de Chile, October 2024.
- Dynamical systems seminar of Santiago, Santiago de Chile, May 2024.
- Conference "Smooth manifolds and related topics", Luxembourg, March 2024.
- Geometry seminar, Université de Rennes, February 2024.
- Geometry seminar, Université de Toulouse, January 2024.
- Geometry seminar, Max Planck Institute, Leipzig, January 2024.
- Winter school in dynamical systems, Aussois, December 2023.
- PhD seminar of the Institut Montpellierain Alexander Grothendieck, Montpellier, November 2023.
- Geometry seminar of the Institut Montpellierain Alexander Grothendieck, Montpellier, November 2023.
- Geometry seminar of the Universität des Saarlandes, Sarrebruck, July 2023.
- Interactions between geometry and topology, to celebrate Ivan Babenko, Banyuls-sur-Mer, June 2023.
- PhD seminar of the Institut Fourier, Grenoble, November 2022.
- PhD days of the Institut Fourier, Grenoble, October 2022.

¹<https://mayaa.world>

²<https://pesu.ens.fr/talens/>

- Workshop "Hyperbolic dynamical systems, geometry and spectral theory", Peyresq, June 2022.
- Conference "Geometry, topology and dynamics in low dimension", Sète, May 2022.
- Number theory seminar of Oregon State University. March 2022.
- Geometry seminar of the Institut mathématique de Marseille. February 2022.
- Winter school "Billards dans le Vercors". January 2022.
- Presentation aimed at a wide audience "*Polygonal billiards and translation surfaces*", as part of the scientific conferences organized by the French Embassy in China. December 2021.
- Geometry and Topology seminar, Institut Fourier, September 2021.

ATTENDED CONFERENCES

- Workshop of the ANR Adyct : "Aléatoire, Dynamique, spectre" in Peyresq, France, July 2025.
- JHH80 Dynamical Developments: Degenerations of Flat Surfaces and Rational Maps. Toulouse, June 2025.
- Geometry and dynamics in dimension 1,2 and 3, Marseille, June 2025.
- Conference "Dyadisc 7: Brazilian, Chilean and French interplay for symbolic dynamics", Valparaiso, December 2024.
- Conference "Smooth manifolds and related topics", Luxembourg, March 2024.
- Winter school in dynamical systems, Aussois, December 2023.
- Conference Renormalization and visualisation in dynamics, geometry and number theory, CIRM, Décembre 2023.
- Meeting for young researchers of the GDR Platon, Grenoble, Novembre 2023.
- Workshop on Renormalization and visualisation in dynamics, geometry and number theory, CIRM, September 2023.
- Reshsearch school on Renormalization and visualisation in dynamics, geometry and number theory, CIRM, July 2023.
- Interactions between geometry and topology, to celebrate Ivan Babenko, Banyuls-sur-Mer, June 2023.
- Winter school in dynamics and geometry, Lille, March 2023.
- Meeting for young researchers of the GDR Platon, Orsay, Novembre 2022.
- Conference "Big mapping class groups", CIRM, October 2022.
- Conference "Combinatorics, Dynamics and Geometry on Moduli Spaces", CIRM, September 2022.
- Workshop "Hyperbolic dynamical systems, geometry and spectral theory", Peyresq, June 2022.
- Sage days 112.5, Montpellier, June 2022.
- Conference "Geometry, topology and dynamics in low dimension", Sète, May 2022.
- Meeting of the ANR Modiff, Bordeaux, April 2022.
- Winder school "Billards dans le Vercors", Autrans, January 2022.
- Winter school in dynamics and geometry, Aussois, December 2021.
- Meeting for young researchers of the GDR Platon, CIRM, November 2021.
- Meeting of the ANR Adyct, Nantes, November 2021.

OTHER

Languages

French (native), english (fluent), spanish (advanced)